

NNMClub Login Failure – Root-Cause Diagnosis

Revision: 1 **Last modified:** 2026-06-16T00:00:00Z **Run-id:** nnmclub-login-diagnosis-20260616
Authority: Â§11.4.6 (no-guessing, FACT-grade), Â§11.4.114 (working-example comparison), Â§11.4.123 (rock-solid proof) **Scope:** READ-ONLY investigation. No production edits, no commit. No credential values captured/printed.

1. Verdict (FACT-grade)

Root cause: nnmclub.to gates login.php behind a Cloudflare Turnstile JS challenge. The Boba password-login POST (_nnmclub_login in download-proxy/src/merge_service/search.py, and the plugin fallback _password_login in plugins/nnmclub.py) sends only username / password / login. The server requires a valid cf-turnstile-response token (produced by solving the JS widget in a real browser) before it will issue the phpbb2mysql_4_sid session cookie. Without it the POST returns **HTTP 200 with NO Set-Cookie** – no session – has_session=False.

Classification: (a) CAPTCHA – operator must supply browser-exported NNMCLUB_COOKIES. This is NOT a code bug in the POST body, NOT a UA/Cloudflare-block on requests (plain curl reaches the form fine), and NOT (provably) an invalid-credentials problem – credentials never get a chance to be evaluated because the Turnstile gate fails first. The dead-domain fix (f6b2761, nnm-club.me – nnmclub.to) correctly restored reachability; the remaining failure is the Turnstile login gate.

2. Captured evidence (from nezha.local – direct internet, no Mac SSH tunnel)

All probes run as `ssh milosvasic@nezha.local 'curl ...'`.

2a. Login form requires a JS CAPTCHA + hidden fields

GET `https://nnmclub.to/forum/login.php:`

```
<form action="login.php" method="post" target="_top" id="loginFrm">
<input name="f" type="hidden" value="-1">
<input type="hidden" name="redirect" value="" />
<input type="hidden" name="code" value="5cab14e06b8e112b
... CAPTCHA ...
<noscript><b>Ð"Ð»Ñ  Ð¾Ñ, Ð¾Ð±ÑÐ°Ð¶ÐµÐ½Ð, Ðµ CAPTCHA, Ð¿Ð¾Ð¶Ð°Ð¶Ð°Ð¶»ÑfÐ¹ÑÑ, Ð°,

```

CAPTCHA widget tokens present in the page:

```
cf-turnstile
turnstile
data-sitekey
```

â†’ **Cloudflare Turnstile.** The visible CAPTCHA only renders with JavaScript; a non-browser client cannot produce the `cf-turnstile-response` token.

2b. The hidden code field is a STATIC anti-CSRF token, not the blocker

Two separate GETs returned the **identical** value `code=5cab14e06b8e112b`. A per-session nonce would differ. So including code would not fix login â€” Turnstile is the gate.

2c. Credential POST (with hidden code included) yields NO session cookie

```
fresh code=5cab14e06b8e112b
POST .../login.php (username=probe_xyz, password=probe_xyz, login=Ð²Ñ...Ð¼Ð
HTTP/2 200
# (no Set-Cookie line in response headers)
```

A successful login would carry `Set-Cookie: phpbb2mysql_4_sid=...`. It is absent â†’ server rejected the login at the Turnstile gate before evaluating credentials.

2d. Cloudflare in front, but NOT blocking the request

```
server: cloudflare
cf-cache-status: DYNAMIC
HTTP/2 200
```

The page loads (HTTP 200), so this is **not** a UA/IP Cloudflare challenge on the request itself â€” option (c) is ruled out. The block is the in-page Turnstile widget the login flow must satisfy.

3. Working-tracker comparison (Â§11.4.114)

Tracker	login.php Turnstile?	Boba auth method	has_session
rutracker.org	none (probe returned empty)	password POST (login_username/ login_password/ login)	True
kinozal	none observed	password POST	True
iptorrents	n/a (cookies)	â€”	True
nnmclub.to	cf-turnstile present	password POST (username/password/ login) â€” Turnstile token missing	False

The Boba `_nnmclub_login` is a faithful clone of the rutracker pattern. It works for rutracker precisely because rutrackerâ€™s `login.php` has **no Turnstile**. `nnmclub.to`â€™s does, so the identical pattern cannot succeed there. The code is not â€œwrongâ€” the targetâ€™s auth surface is fundamentally different (browser-only).

Probe contrast:

```
rutracker login.php turnstile grep â†’ (empty) # no widget
nnmclub login.php turnstile grep â†’ cf-turnstile / turnstile / data-site
```

4. Why the existing diag message is misleading (Â§11.4.6)

search.py:1538-1547 records on failure:

```
"nmclub login returned no session cookie â€” likely credential failure"
```

The word **likely** is a Â§11.4.6 no-guessing violation, and the guess is wrong: the FACT is a Turnstile gate, not a credential failure. Suggested corrected wording: "nmclub_login.php is gated by Cloudflare Turnstile (JS CAPTCHA); password POST cannot obtain a session â€” set NMCLUB_COOKIES from a browser session."

5. Classification & required action

Classification (a): CAPTCHA â†’ browser-exported cookies (operator/env, Â§11.4.3 topology).

Operator action (the working path â€” already supported by code)

_search_nmclub (search.py:1466-1474) ALREADY prefers NMCLUB_COOKIES when set and skips the password login entirely. The operator must export cookies from a logged-in browser session at nmclub.to and set the NMCLUB_COOKIES env var.

Exact cookies required (the codeâ€™s hard gate is phpbb2mysql_4_sid; a full working session also carries the data + persistence cookies â€” export all phpbb2mysql_4_*): - phpbb2mysql_4_sid â†’ **REQUIRED** â€” search.py:1479 / plugin login() line 212 reject without it - phpbb2mysql_4_data â†’ needed for persistent auth (plugin_init line 341 keys on this) - cf_clearance â†’ recommended â€” Cloudflare clearance cookie, lets subsequent requests bypass the Turnstile interstitial

NMCLUB_COOKIES format (semicolon-separated, per search.py:1470 and plugin login() line 188):

```
NMCLUB_COOKIES="phpbb2mysql_4_sid=<...>; phpbb2mysql_4_data=<...>; cf_clearance=<...>"
```

Cookies expire periodically (same operational caveat as RuTracker per CLAUDE.md) and must be re-exported when nmclub auth lapses.

NOT recommended (out of scope / fragile)

Automating Turnstile solving (3rd-party solver services, headless-browser token mint) is brittle, may violate the tracker ToS, and is not the established Boba pattern. The cookie-injection path (Â§11.4.3 operator-supplied evidence) is the correct, robust fix.

No code fix needed for login to work

The cookie path already works end-to-end in code. The only code-quality follow-ups (separate, optional): 1. Fix the Â§11.4.6 **likely** credential failureâ€™ diag string (section 4) â†’ state the Turnstile FACT so operators know to supply cookies. 2. Optionally have _nmclub_login detect cf_turnstile in the login page and emit an error_type:

"upstream_captcha" diag (mirroring rutracker's upstream_captcha label at search.py:1255-1257) instead of auth_failure.

RED-test design (for the diag-string follow-up, if pursued)

- RED: assert that when _nnmclub_login is driven against a login page containing cf -turnstile, the recorded diag_error_type == "upstream_captcha" (currently auth_failure) and the message contains "Turnstile", not "likely". Fails pre-fix.
 - GREEN: classifier detects the cf -turnstile marker; flip RED_MODE=0 as the guard.
-

6. Files referenced

- /Volumes/T7/Projects/boba/download-proxy/src/merge_service/search.py " _search_nnmclub (1454), _nnmclub_login (1501), diag (1538), cookie-precedence (1466)
- /Volumes/T7/Projects/boba/plugins/nnmclub.py " login() (180), _password_login() (217), _init() cookie load (311)
- /Volumes/T7/Projects/boba/download-proxy/src/api/auth.py " nnmclub auth_status branch (not the cause; reflects has_session truthfully)